Math Inquirers

# Extension Workbook

## with answers

**Level 3**

Self-study | Ages 8-10




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 3**
Self-study practice with answers.


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## Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

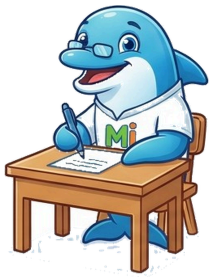

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# Welcome!




This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

## How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

## For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- You will need: a pencil, a rubber, a ruler and a paper corner.

## The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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| 2 | **Cryptarithm crackers**<br>Not in this proof edition | – |
| 3 | **Times table treasure**<br>Not in this proof edition | – |
| 4 | **Model makers (bar models)**<br>Not in this proof edition | – |
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# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | Number sleuths (numbers to 10 000) | Not in this proof edition | | | | ☐ |
| 2 | Cryptarithm crackers | Not in this proof edition | | | | ☐ |
| 3 | Times table treasure | Not in this proof edition | | | | ☐ |
| 4 | Model makers (bar models) | Not in this proof edition | | | | ☐ |
| 5 | Remainder riddles | Not in this proof edition | | | | ☐ |
| 6 | Fraction walls | Not in this proof edition | | | | ☐ |
| 7 | Area architects | Not in this proof edition | | | | ☐ |
| 8 | Angle and line detectives | Not in this proof edition | | | | ☐ |
| 9 | Measure and money masters | Not in this proof edition | | | | ☐ |
| 10 | Think it through | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 11 | Catch-up week | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

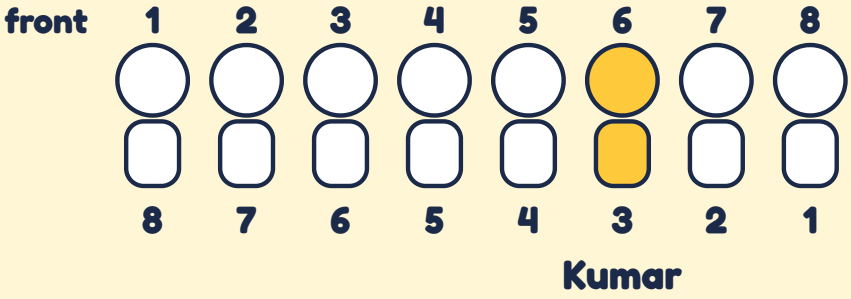
# Learn it: Mixed puzzles

**My goal.** I can start a puzzle by drawing it, listing it or making a table. I can read a bar graph and solve logic puzzles.

To start a puzzle, try one of 3 ways. Draw it, list it, or make a table. Then check every clue.

## Worked example

**Kumar is 6th from the front and 3rd from the back of a line. Count the children in the whole line.**




1. Draw the line. Put Kumar in 6th place from the front.
2. Kumar is 3rd from the back, so 2 children are behind him.
3. 5 children are in front of Kumar, then Kumar, then 2 behind: 5 + 1 + 2 = 8.
4. Check: 6 + 3 − 1 = 8.

**Answer: There are 8 children in the line.**

## Three ways to start

1. Draw it: make a picture of the puzzle.
2. List it: write every case in order.
3. Table: write what you know in rows and columns.
4. Check your answer against every clue.

## Your turn

Answer at the back of the book.

**Try** Lily is 2nd from the front and 7th from the back of a line. The line has ☐ children.

## Start here

Do these first. Take your time.

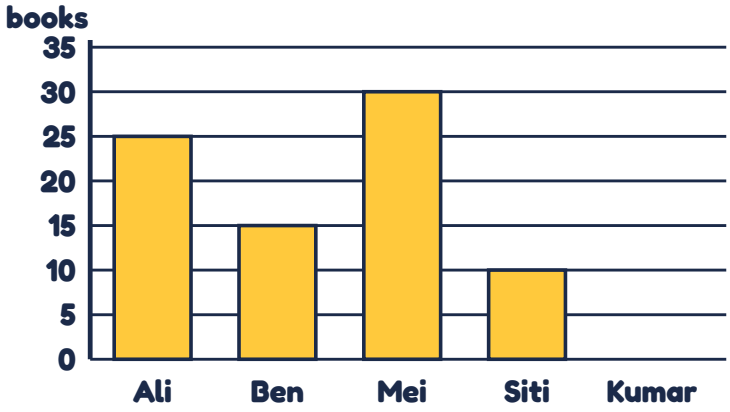
**1** **The bar graph shows the books 4 children read. Kumar read 15 more books than Siti. Draw Kumar's bar on the graph.**

(a) Most books: ☐

(b) Mei read ☐ more than Ben.

(c) Kumar read ☐.

(d) In all they read ☐.



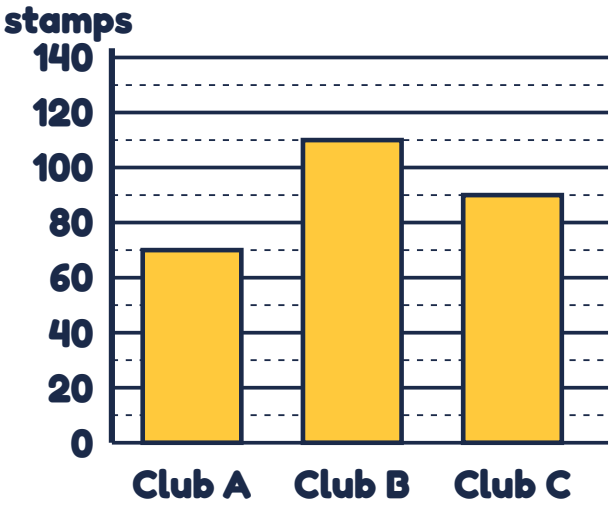

**2** **The graph shows the stamps 3 clubs collected. Each thin line is 10 stamps.**

(a) Club A: ☐

(b) Club B: ☐

(c) Club C: ☐

(d) All 3: ☐




**3** **The 3rd of a month is a Tuesday.**

(a) The 24th is a Mon Tue Wed Thu Fri Sat Sun

(b) The 30th is a Mon Tue Wed Thu Fri Sat Sun

Working: _____

## Start here

Keep going. Check your work.

**4** **Each child has a different fruit. Kumar has neither the apple nor the banana. The child with the orange has a name with 4 letters. Wei does not have the apple. Use ticks and crosses in the grid.**

| | apple | banana | grapes | orange |
|---|---|---|---|---|
| Kumar | | | | |
| Aisha | | | | |
| Wei | | | | |
| Lily | | | | |

(a) Kumar has the apple banana grapes orange

(b) Aisha has the apple banana grapes orange

(c) Wei has the apple banana grapes orange

(d) Lily has the apple banana grapes orange

**5** **Draw it. Count the children in each line.**

(a) Ravi is 3rd from the front. Mei is last. 5 children stand between them. The line has ☐ children.

(b) Aisha is 8th from the front and 2nd from the back. The line has ☐ children.

Draw it:

**6** **Make every 3-digit number from the cards 3, 6 and 9. Use each card once in a number. List them from the smallest.**

(a) Write the list below.

(b) There are ☐ even numbers in your list.

(c) The number closest to 700 is ☐.

My list: _____

## Start here

Nearly there. You can do it.

**7** **Hexagons are joined in a row with sticks. 1 hexagon needs 6 sticks. 2 hexagons need 11 sticks. 3 hexagons need 16 sticks. Make a table.**



(a) 6 hexagons need ☐ sticks.

(b) 41 sticks make ☐ hexagons.

| hexagons | sticks |
|---|---|
| | |
| | |
| | |
| | |

**8** **Which numbers fit all the clues? The number is even and has 3 digits. It is less than 400. Its digits add up to 9, and none is 0. The tens digit is more than the ones digit.**

_____

**9** **I think of a number. I multiply it by 3 and take away 4. The answer is 17. My number is ☐.**

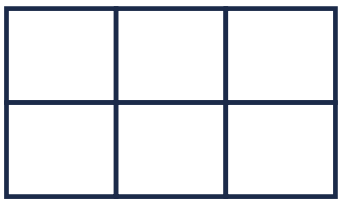
Working backwards: _____

**10** **Aisha has $10. A juice costs $1.85 and a sandwich costs $3.40. She buys 2 juices and 1 sandwich. Her change is $ ☐.**

## Think harder

Hints are upside down. Try first!

**1** **A grid is made of 2 rows of 3 small squares. Count the rectangles of every size, squares included.** ☐



1 by 1: 1 by 2: 1 by 3: _____

2 by 1: 2 by 2: 2 by 3: _____

**Hint.** Count 1 by 1 first. Then 1 by 2. Make a table.

**2** **One of Ben, Aisha and Wei broke a cup. They each say 1 thing: Ben "Wei did it", Aisha "Not me", Wei "Ben is lying". Exactly 1 of them tells the truth. Who broke the cup? Ben Aisha Wei**

My tests: _____

_____

_____

**Hint.** Pick one child. Are exactly 1 of the 3 sayings true?

**3** **A shop sold 185 pencils in 4 days. Tuesday sold 15 more than Monday. Wednesday sold twice as many as Monday. Thursday sold 10 fewer than Wednesday.**

(a) Monday: ☐ (b) Tuesday: ☐ (c) Wednesday: ☐

(d) Thursday: ☐

Draw your bars:

_____

**Hint.** Draw Monday as 1 bar. Then draw the other days with it.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Count the squares of every size in a 4 by 4 grid. Begin with the small ones.**

(a) 1 by 1 squares: ☐ (b) 2 by 2 squares: ☐

(c) 3 by 3 squares: ☐ (d) All the squares: ☐

**Hint.** A 2 by 2 square can sit in 3 places across and 3 down.

★ Beyond Level 3: A balance puzzle

**2** **9 coins look the same. One is lighter. You have a balance scale.**

(a) With 3 coins, you can always find the light coin in ☐ weighing.

(b) With 9 coins, you can always find it in ☐ weighings.

**Hint.** Put 3 coins on each side. What does the scale tell you?

**3** **Ali has 8 coins. They are 10¢ coins and 20¢ coins, worth $1.50 in all. Find how many coins of each kind.**

(a) ☐ ten-cent coins

(b) ☐ twenty-cent coins

| 10¢ coins | 20¢ coins | value |
|---|---|---|
| | | |
| | | |

**Hint.** Try 4 of each. How much are they worth?

**4** **Make up a logic puzzle with 3 friends and 3 sports. Write 3 clues. Only ONE answer may fit. Swap with a grown-up.**

My clues: _____

_____

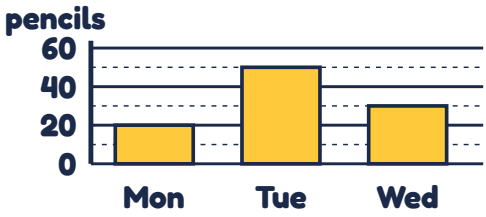
_____

**Hint.** Pick the answer first. Then write clues that rule out the others.

## Check yourself

No hints. Try each one on your own.

**1** The graph shows pencils sold. Tuesday sold ☐ more than Monday.




**2** In a line, 3 children stand in front of Ben and 4 children stand behind him. The line has ☐ children.

Working: _____

**3** Use the cards 2, 5 and 6 once each to make 3-digit numbers. ☐ of the numbers are odd.

Working: _____

**4** Look for a pattern. 2, 6, 10, 14, … The 8th number is ☐.

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Number sleuths (numbers to 10 000) | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Cryptarithm crackers | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Times table treasure | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Model makers (bar models) | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | Remainder riddles | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Fraction walls | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Area architects | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Angle and line detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | Measure and money masters | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Think it through | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

**You finished the Math Inquirers Extension Workbook, Level 3.**

Date

Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 10 · Think it through

**Your turn**
page 6

**Answer:** 8
**Why:** 1 child is in front of Lily and 6 are behind her. 1 + 1 + 6 = 8. Check: 2 + 7 - 1 = 8.

## START HERE

**1**
page 7

**Answer: (a) Mei (b) 15 (c) 25 (d) 105**
**Why:** (a) The tallest bar is Mei's. (b) 30 - 15 = 15. (c) Siti read 10 books, so Kumar read 10 + 15 = 25. (d) 25 + 15 + 30 + 10 + 25 = 105.

**2**
page 7

**Answer: (a) 70 (b) 110 (c) 90 (d) 270**
**Why:** The labelled lines go up in 20s and a thin line is halfway between, so each thin line is 10. A is 70, B is 110, C is 90. Together: 70 + 110 + 90 = 270.

**3**
page 7

**Answer: (a) Tue (b) Mon**
**Why:** (a) 3rd to 24th is 21 days = 3 weeks. So the 24th is a Tuesday again. (b) 3rd to 30th is 27 days = 3 weeks and 6 days. 6 days after Tuesday is Monday.

**4**
page 8

**Answer: (a) grapes (b) apple (c) banana (d) orange**
**Why:** Only Lily has a 4-letter name: orange. Kumar has neither apple nor banana: grapes. Wei does not have the apple: banana. Aisha has the apple.

**5**
page 8

**Answer: (a) 9 (b) 9**
**Why:** (a) Ravi is 3rd. The 5 children between are 4th to 8th. Mei is 9th and last. (b) 7 children are in front of Aisha and 1 is behind her: 7 + 1 + 1 = 9.

**6**
page 8

**Answer: (a) 369, 396, 639, 693, 936, 963 (b) 2 (c) 693**
**Why:** Fix the first digit and list the rest in order: 369, 396, then 639, 693, then 936, 963. The even numbers end in 6: 396 and 936. 693 is only 7 from 700; the others are much farther.

**7**
page 9

**Answer: (a) 31 (b) 8**
**Why:** Each new hexagon adds 5 sticks: 1: 6, 2: 11, 3: 16, 4: 21, 5: 26, 6: 31. For 41 sticks, count on: 7 hexagons need 36 and 8 hexagons need 41.

**8**
page 9

**Answer: 162, 252 and 342**
**Why:** The hundreds digit is 1, 2 or 3. The ones digit is 2, 4 or 6 and the tens digit is greater. With 1: tens + ones = 8, so 6 and 2. With 2: 7, so 5 and 2. With 3: 6, so 4 and 2. The numbers are 162, 252 and 342.

**9**
page 9

**Answer: 7**
**Why:** Work backwards. Add 4 to 17: 21. Divide by 3: 7.

**10**
page 9

**Answer: 2.90**
**Why:** 2 juices cost 2 × 1.85 = 3.70. With the sandwich: 3.70 + 3.40 = 7.10. Change: 10.00 - 7.10 = 2.90.

## THINK HARDER

**1** page 10

**Answer: 18**
**Why:** 1 by 1: 6. 1 by 2 (2 wide, 1 tall): 2 in each row, so 4. 1 by 3: 2. 2 by 1 (1 wide, 2 tall): 3. 2 by 2: 2. 2 by 3: 1. In all: 6 + 4 + 2 + 3 + 2 + 1 = 18.

**2** page 10

**Answer: Aisha**
**Why:** If Ben did it, Aisha and Wei both tell the truth: 2 truths. If Wei did it, Ben and Aisha both tell the truth: 2 truths. If Aisha did it, only Wei tells the truth: exactly 1. So Aisha did it.

**3** page 10

**Answer: (a) 30 (b) 45 (c) 60 (d) 50**
**Why:** Monday is 1 box, Tuesday 1 box and 15, Wednesday 2 boxes, Thursday 2 boxes take away 10. Total: 6 boxes + 5 = 185, so 1 box = 30. Monday 30, Tuesday 45, Wednesday 60, Thursday 50.

## GO FURTHER

**1** page 11

**Answer: (a) 16 (b) 9 (c) 4 (d) 30**
**Why:** There are 16 small squares. A 2 by 2 square fits in 3 places across and 3 down: 9. A 3 by 3 square fits 2 across and 2 down: 4. There is 1 big square. 16 + 9 + 4 + 1 = 30.

**2** page 11

**Answer: (a) 1 (b) 2**
**Why:** (a) Put 1 coin on each side. If the scale tips, the light coin is on the high side. If it balances, it is the coin left out: 1 weighing. (b) Weigh 3 coins against 3 coins to find the group of 3 with the light coin. Then use (a): 2 weighings.

**3** page 11

**Answer: (a) 1 (b) 7**
**Why:** If all 8 coins were 10¢, they would be worth 80¢. They are worth 150¢, 70¢ more. Changing one 10¢ coin to a 20¢ coin adds 10¢, so 7 coins are 20¢ and 1 coin is 10¢.

**4** page 11

**Answer: Answers vary. Example: Lily, Wei and Kumar each play a different game: chess, ball or cards. Lily does not play ball. Wei plays neither chess nor cards. Kumar does not play chess. Answer: Lily plays chess, Wei plays ball, Kumar plays cards.**
**Why:** In the example, Wei plays ball (clue 2), then Kumar cannot play chess (clue 3), so Kumar plays cards and Lily plays chess.

## CHECK YOURSELF

**1** page 12

**Answer: 30**
**Why:** Monday is at 20 and Tuesday is at 50. 50 - 20 = 30.

**2** page 12

**Answer: 8**
**Why:** 3 + 1 + 4 = 8. Do not forget Ben.

**3** page 12

**Answer: 2**
**Why:** The numbers are 256, 265, 526, 562, 625 and 652. The odd ones end in 5: 265 and 625.

**4** page 12

**Answer:** 30
**Why:** The numbers go up by 4. The 8th number is 2 + 7 × 4 = 30.